Closure or Information? Weekend Variance and Option Prices in a Market That Never Closes

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Abstract

Asset prices are far more volatile when exchanges are open than when they are shut, but the cause has been difficult to establish because closure and the absence of trading are the same event in equity markets. We use a setting in which they are not. Deribit trades cryptocurrency options and perpetual futures continuously, including weekends, while the traditional financial system does not. We document that realized variance is 34% to 42% lower on weekends than on weekdays across four underlyings, with Saturday the quietest day of the week in all of them, and that the effect roughly doubles for tokenized gold, whose own market in London and on COMEX is shut while its Deribit contract keeps trading. We then ask whether options price what we measure. Because the exchange lists daily expiries on all seven weekdays, contracts quoted at the same instant differ in how much weekend calendar time they span, and we use that variation to identify the implied weekend discount within the trading day. We find that the market prices roughly seven eighths of the effect. Its one demonstrable failure lies within the weekend rather than across assets: every book prices Saturday as indistinguishable from Sunday when Saturday is reliably the quieter day. We show that the exchange's own expiry schedule makes this particular error unarbitrageable, since the two contracts needed to spread it against each other coexist in only 13 hours of the week and never on a weekday.

Keywords: calendar time; realized variance; option pricing; market microstructure; cryptocurrency derivatives

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1. Introduction

French and Roll (1986) established that variance per calendar hour collapses when an exchange is shut, and proposed three explanations: public information arrives disproportionately during business hours, private information is impounded when informed investors trade, or trading itself generates pricing errors. Distinguishing among them has proved difficult for a structural reason. In a conventional market, the exchange being closed and no one trading are the same event, and both coincide with the hours in which most public information is released. The three explanations are confounded by construction.

We study a market in which they are not. Deribit trades options and perpetual futures continuously, weekends included, while the traditional financial system does not. Because trading remains possible at all times, any weekend variance effect we find cannot be attributed to a venue being shut.

We document three results. The first concerns realized variance. Weekend variance is 42% below weekday variance for Bitcoin, 39% for Ether, 38% for XRP and 34% for Solana, and Saturday is the quietest day of the week in each. We read this as evidence that much of what moves cryptocurrency prices originates in traditional market hours, whether through macroeconomic releases, institutional flow or the spot-ETF complex. The reading is hard to avoid because four assets that differ in settlement convention, holder base and listing history nonetheless inherit the same weekly rhythm. We sharpen it with a fifth asset. PAXG, tokenized gold, trades continuously on the same venue while gold's own market is shut all weekend, and its weekend variance is 65% below weekday, roughly double the cryptocurrency effect. Raising the amount of the traditional calendar that is switched off deepens the weekend, on the same exchange and under the same estimator.

The second result concerns prices. Deribit lists daily expiries on all seven weekdays, so contracts quoted at the same instant differ in how much weekend calendar time they span: on a Thursday, a Friday expiry covers no weekend while a Monday expiry covers most of one. We regress squared implied volatility on that fraction with day fixed effects, which absorb the level of volatility and every other shock common to the day, and identify the implied weekend discount from variation within the trading day alone. No equity market offers this comparison, because no contract can expire while the exchange is closed. We find that Bitcoin's options price 85.0% of Bitcoin's own realized weekend effect and Ether's 85.5% of Ether's. None of the four residual pricing errors is individually significant, and we do not build on them.

The third result concerns where the market goes wrong, and we find that it is within the weekend rather than across assets. Every book prices Saturday as statistically indistinguishable from Sunday, though Saturday is reliably the quieter of the two. Elsewhere in the week the implied day-of-week profile tracks the realized one closely, with correlations of +0.98, +0.88, +0.89 and +0.82, so the market reads the calendar well everywhere except here, where it treats the weekend as a single undifferentiated block. Four independent books make the same mistake in the same place.

We then ask why that error survives, and we argue that the answer lies in the contract calendar rather than in beliefs. Correcting it requires selling the Saturday-heavy contract and buying the Sunday-heavy one, which in turn requires both to exist at the same moment. Enumerating every hour of the week against every listed expiry, we find that they coexist in 13 hours out of 168, none of them on a weekday. From Monday to Friday no Sunday-heavy contract is listed at all, because any contract that reaches Sunday must pass through Saturday first. The one ranking of calendar time the market demonstrably gets wrong is the one its own expiry schedule prevents anyone from arbitraging.

Two further exercises bound how our results should be read. We take the risk-premium explanation seriously and race it. Weekend returns do carry fatter tails, so weekend-specific risk exists and must be priced rather than dismissed, but when we decompose realized variance into continuous and jump components we can bound what any jump premium is able to buy, and every book's quote falls outside that bound. Two of the four would require a negative price of jump risk, because their quotes already sit below their own realized weekend variance. We also ask whether the residual error can be harvested, and find that it cannot: a vega-matched calendar spread isolating it earns +0.042 per unit vega gross against 0.066 of measured fees and spread, so it has never cleared zero at taker cost.

The design point we want to emphasize is that a single venue supplies both halves of the test. On the realized side we hold closure fixed and let the information environment vary, then raise the dose with an asset whose own market shuts. On the implied side we compare contracts quoted in the same book at the same instant that differ only in weekend coverage. Neither comparison is available in equity or listed-futures markets, where closure, non-trading and the news calendar move together. What we buy with it is a measurement of how a derivatives market prices a known, mechanical and repeating feature of calendar time, and, because the same venue supplies the benchmark, a measurement of how well. Our answer is that it prices the level substantially correctly, and fails on resolution and on the choice of moment.

Related literature

French and Roll (1986) is the origin of the question, though Oldfield and Rogalski (1980) had already framed returns as accruing over trading and non-trading periods that need not share a clock. Barclay, Litzenberger and Warner (1990) came closest to separating the mechanisms, using a natural experiment that runs in the opposite direction from ours: when the Tokyo Stock Exchange opened on Saturdays, weekend variance rose while weekly variance was unchanged despite higher volume. They varied whether an exchange was open while holding the information environment roughly fixed, whereas we hold the exchange open always and vary whether the traditional system is running. The two designs bracket the question from opposite sides, and both point to trading time rather than calendar time as the relevant clock.

Practitioners have long used trading-day rather than calendar-day conventions in option pricing, but the academic treatment of weekend and holiday effects in implied volatility remains thin relative to its practical importance. Our contribution on that side is a venue in which the clock convention is identified from variation within the trading day rather than assumed.

Three further literatures bear on how we interpret the residual. Whether it represents compensation rather than error is a variance-risk-premium question in the sense of Carr and Wu (2009), and we answer it in the negative for this particular quantity in Section 6. Whether it instead reflects order flow rather than beliefs is the demand-pressure question of Gârleanu, Pedersen and Poteshman (2009) and Bollen and Whaley (2004). We cannot address that one, because the trade tape carries the aggressor side but not the dealer's inventory, and we prefer to say so rather than infer. Finally, the cryptocurrency option market has developed its own measurement literature, and we draw on Alexander and Imeraj (2023) for the hedging conventions used in Section 7.

2. Institutional setting and data

2.1 Why Deribit

Deribit is the dominant venue for cryptocurrency options and, critically for our design, lists daily expiries on all seven weekdays. Across the 342,827 instruments in the four books we use, the expiry-weekday distribution is close to uniform apart from the expected Friday concentration, with every underlying listing between 12% and 14% of its instruments on each of the six non-Friday weekdays. From 2020 onward an expiry exists on essentially every calendar date, 365 of 365 in each year from

2021 to 2025, and short-dated contracts are the norm rather than a curiosity: roughly 80% of instruments have seven days or less of life.

That schedule is what makes our design work. Because Saturday and Sunday expiries exist and trade, a contract quoted on a Thursday afternoon may span no weekend at all or most of one, and both sit in the same book at the same instant. No equity or listed-futures market offers this, since nothing can expire while the exchange is shut and weekend coverage is therefore collinear with maturity.

2.2 Sample

We use the complete public trade history collected from `history.deribit.com`, comprising 24,349,954 Bitcoin trades from November 2016, 16,207,332 Ether trades from March 2019, and 695,295 Solana and 234,640 XRP trades from early 2024, running to August 2026. That is 41.5 million trades in total. Each carries the exchange's own implied volatility, the index level, the aggressor side, and flags for block trades, combinations and liquidations.

Our baseline sample excludes block trades and combinations, which are bilaterally negotiated or double-counted, retains liquidations, and keeps contracts with $|\Delta|$ between 0.30 and 0.70 and maturity between six hours and fourteen days. We compute realized variance from five-minute perpetual-future returns over 2,923 Bitcoin days, 2,704 Ether days, 915 Solana days and 887 XRP days, at 100%, 100%, 99.9% and 99.9% bar completeness. Five minutes is the conventional compromise between discretization error and microstructure noise (Zhang, Mykland and Aït-Sahalia 2005, Hansen and Lunde 2006), and we vary it from five minutes to two hours in Section 5.3, where the direction in which estimates move under that variation serves as a diagnostic rather than as a robustness check. We drop returns that span a feed gap rather than winsorizing them, since a multi-period return carrying a one-period label would inflate exactly the tail statistics Section 6 depends on.

Solana and XRP options are USDC-settled and linear rather than coin-settled, and they carry contract sizes of ten and one thousand rather than one. We therefore apply separate premium and hedging conventions throughout, which is what makes them genuinely independent replications rather than relabellings of the same book. One caveat attaches to both. We use no forward curve for either, because Deribit's dated futures on these underlyings are close to untraded and only three SOL and three XRP contracts produced usable daily closes over the whole sample. We compute their greeks against the index instead, and document the gap rather than patch it.

2.3 The reference asset

A fifth asset enters on the realized side only. PAXG is tokenized gold: its Deribit perpetual trades continuously like every other book here, but the market that forms gold's price, in London and on COMEX, is shut from Friday 22:00 to Sunday 22:00 UTC. It therefore raises the dose of our treatment rather than merely exhibiting it, and we use it in Section 3 as a dose-response check.

We take no pricing estimate from it. Of its 5,346 listed options, 5,304 expire on a Friday, so within a trade day the weekend fraction of a contract's remaining life is a deterministic function of its maturity and cannot be separated from the maturity controls that Section 5.1 requires. Its perpetual covers 618 days from December 2024.

3. The fact: weekends are quiet in a market that is open

Table 1 reports annualized realized volatility by day of week together with the implied weekend-to-weekday variance ratios.

Table 1. Realized volatility by weekday, and weekend variance ratios.

	Mon	Tue	Wed	Thu	Fri	**Sat**	**Sun**	Weekend/weekday variance
BTC	59.9	58.9	61.1	60.4	59.2	**42.0**	**46.3**	**0.584**
ETH	77.0	75.9	79.2	77.1	75.3	**56.7**	**62.4**	**0.607**
SOL	89.0	85.5	83.6	82.0	87.6	**60.8**	**68.8**	**0.657**
XRP	86.5	80.1	76.5	76.0	84.2	**58.3**	**63.9**	**0.621**

Annualized percent, from five-minute returns. Weekday and weekend day counts are 2,089/834 (BTC), 1,932/772 (ETH), 655/260 (SOL) and 634/253 (XRP).

Saturday is the quietest day of the week in every asset and Sunday the second quietest, and no weekday in any row falls below any weekend day. Weekend variance runs 34% to 42% below weekday variance in all four books while the exchange remains open throughout.

We regard the agreement across assets as a sharper test of the mechanism than the level of any single ratio, and it is worth being explicit about why. If cryptocurrency weekends are quiet because the traditional financial system is closed, then every cryptocurrency asset should inherit the same weekly rhythm regardless of its own microstructure, since what stops on Saturday is external to all of them.

If instead weekend quiet reflected asset-specific features such as retail participation, venue depth or the composition of the holder base, we would expect the ratios to diverge, because those features differ sharply across these four. The ratios agree within seven percentage points despite differences in settlement convention, contract size, listing history and volatility level.

The alternative explanation, that trading itself generates variance and weekends are quiet because volume is lower, is not available here in the form it takes in equity markets. The venue is open, the book is live, and the perpetual trades throughout. It survives only as a claim about endogenous participation, namely that traders choose not to trade at weekends, which is itself most naturally explained by there being less to trade on.

We turn the traditional calendar up using the reference asset. All four cryptocurrency books sit at the same dose of the treatment, in that their own venues never close while the system they take information from is shut every weekend. PAXG raises it, because the weekend removes not merely the flow of macroeconomic news but the price-formation process itself.

PAXG's weekend variance ratio is 0.347, against 0.584 to 0.657 for the four cryptocurrency assets, which is a weekend discount of 65% where theirs is 34% to 42%. Venue, estimator, sampling frequency and day-type definition are identical, and the only thing that changes is whether the underlying's own market is running. We find a dose-response of this shape difficult to produce from any account that locates the effect in cryptocurrency microstructure.

We report this as a lower bound, and the reason matters for how it should be read. PAXG's perpetual repeats 88% of its weekend five-minute closes against 78% of its weekday ones, and stale prices depress measured variance in both regimes while biasing their ratio toward one. The true weekend effect in gold is therefore at least as large as we report. Its variance ratio also falls monotonically, from 0.345 to 0.188, as the sampling interval coarsens and the staleness is removed, which is the signature of exactly this bias. We find no such drift in the four traded books, and that is what licenses reading their ratios at face value.

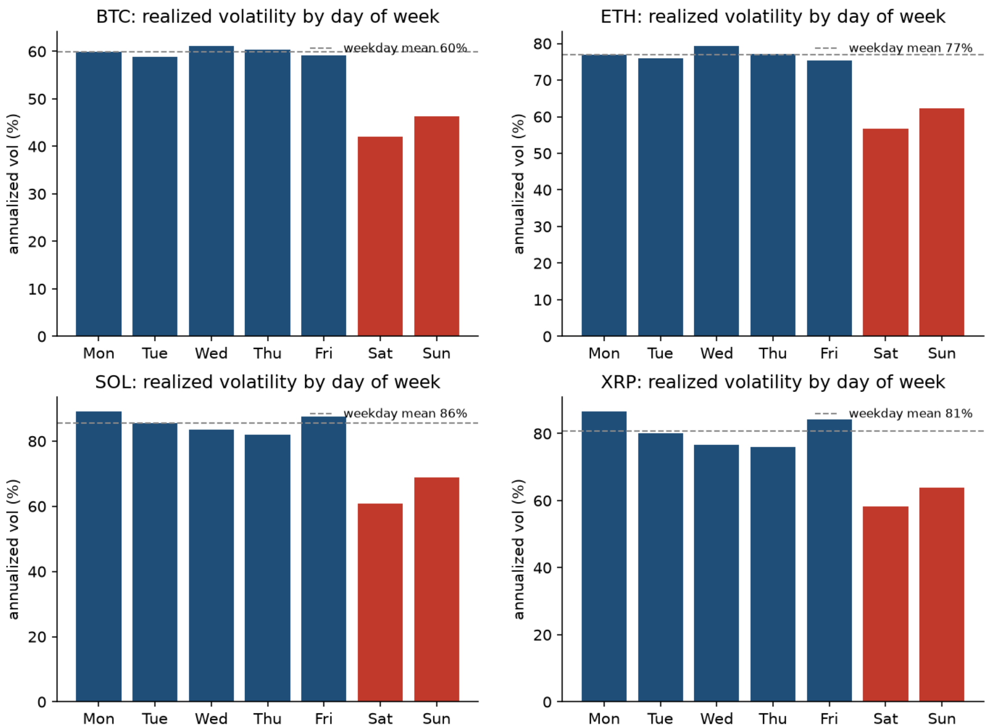

Figure 1. Realized volatility by day of week. Annualized, from five-minute returns, as a deviation from each asset's own weekday mean. Saturday and Sunday shaded.

4. Identification

We write the total variance over an option's life as a time-weighted average of a weekday and a weekend variance:

$$\sigma_{i,t}^2 T_{i,t} = v_t^{wd}(1 - w_{i,t})T_{i,t} + v_t^{we}\, w_{i,t} T_{i,t},$$

where $w_{i,t}$ is the fraction of contract i's remaining life at time t that falls on a Saturday or Sunday. Dividing through gives

$$\sigma_{i,t}^2 = v_t^{wd} + (v_t^{we} - v_t^{wd})\, w_{i,t},$$

and we estimate

$$\sigma^2_{i,t} = \gamma_t + \beta\, w_{i,t} + \delta' X_{i,t} + \varepsilon_{i,t},$$

with γ_t a day fixed effect and X containing log maturity, its square, and $|\Delta|$. The coefficient β estimates $v^{we} - v^{wd}$, and we report the implied weekend variance ratio $(\bar{v}^{wd} + \beta)/\bar{v}^{wd}$. We cluster standard errors by day throughout.

Two features make us confident in this specification. The day fixed effect absorbs the level of volatility and everything else common to the day, including the state of the market, the funding environment and any news, so that β is identified only from contracts quoted at the same instant that differ in weekend exposure. And w is mechanical, being a property of the calendar and the expiry date rather than of anything the market chooses in response to volatility. A contract's weekend fraction was determined when the exchange set its listing schedule, typically years earlier.

We compute weekend fraction exactly, in closed form, rather than by counting whole days. Deribit expiries settle at 08:00 UTC, so almost every contract covers part-days at both ends, and a whole-day count would misstate w by up to two thirds of a day on precisely the short contracts that carry most of the identifying variation. The same construction yields the fraction falling on each of the seven weekdays, which we use in Section 5.2.

The maturity controls deserve comment, because w and T are mechanically related: a very short contract listed on a Friday is nearly all weekend, while a long one sits close to the 2/7 calendar share. If the implied volatility surface has any term structure, and it does, then omitting maturity would load part of that term structure onto β. Log maturity and its square absorb the smooth part, and the identifying variation that remains is the difference between two contracts of similar maturity, quoted at the same instant, whose expiry dates fall on different weekdays. That variation exists in this venue precisely because expiries are daily.

The threat our design does not rule out by construction is a listing schedule chosen in response to expected weekend variance. We treat its absence as a maintained assumption. A calendar that lists an expiry on all 365 days of the year, fixed in advance and uniform across underlyings, makes it a mild one.

5. Results

5.1 The market prices most of the weekend effect

Table 2. Implied and realized weekend variance ratios.

	slope β	se	t	n	implied ratio	realized ratio	gap	(se)
BTC	−0.1612	0.0149	**−10.8**	5,339,133	0.635	0.584	+0.051	(0.066)
ETH	−0.2461	0.0277	**−8.9**	3,624,938	0.645	0.607	+0.038	(0.067)
SOL	−0.3568	0.0243	**−14.7**	189,906	0.558	0.657	−0.099	(0.102)
XRP	−0.5513	0.0691	**−8.0**	70,128	0.488	0.621	−0.133	(0.123)

Standard errors clustered by day, over 3,413, 2,703, 886 and 885 days. The gap's standard error combines the implied slope's with the realized ratio's own sampling error.

The slope is negative and overwhelmingly significant in all four books, so the market does price weekend calendar time and prices a large share of it. Bitcoin implies a weekend variance ratio of 0.635 against a realized 0.584, which is a discount of 36.5% where the true discount is 41.6%, and Ether implies 0.645 against 0.607. Roughly seven eighths of the effect is in the price in both established books.

We emphasize that the residual gaps are not individually significant, and this constrains everything we say about them. The four t-statistics are +0.78, +0.57, −0.92 and −1.07. We estimate the implied side to within about three points of variance ratio, but the realized side is a difference of two means over at most 834 weekend days, and it dominates the uncertainty. Nothing below should be read as a measured pricing error in a particular book.

The identification is visible in the data without any regression. When we demean both squared implied volatility and weekend fraction within each trading day, we strip out the volatility level and every common shock and are left with the comparison the regression uses. Binned that way, the relationship is close to linear across the full range of weekend fraction in all four books, which is what the constant-within-regime variance assumption behind Section 4 requires. We report the binscatters in the e-companion.

The pattern also survives a cruder cut. Average at-the-money implied volatility for contracts with three days or less to expiry, sorted by expiry weekday, puts Sunday expiries cheapest of the week in all four books and Monday expiries second-cheapest in three, which are precisely the two that carry the most weekend calendar time. The market's ordering is right, and its magnitude is what Table 2 measures.

We next ask whether the market applies one discount to every underlying or calibrates to each. Pooling the four books and testing the restrictions directly, we reject a strictly uniform discount. Beyond that the test is close to uninformative, and we think saying so is more useful than reporting the non-rejection of the alternative.

The dispersion shows why. Across the four assets the standard deviation of implied weekend discounts is 0.0919 against 0.0384 for realized ones, so the market's quotes vary 2.4 times more across assets than the weekend risk they are pricing does. We read that as over-differentiation rather than calibration. The proportionality hypothesis nonetheless fails to reject, and the reason is power rather than support: the four realized effects are bunched inside a span of 0.092, so dividing by them is close to dividing all four slopes by the same constant, and the ratio test becomes nearly the equality test. It survives only because the delta-method correction for realized-effect uncertainty inflates the standard errors on the two short samples to 0.418 and 0.502, against 0.178 and 0.187 for the mature books. A non-rejection obtained that way is a statement about the sample rather than about the market.

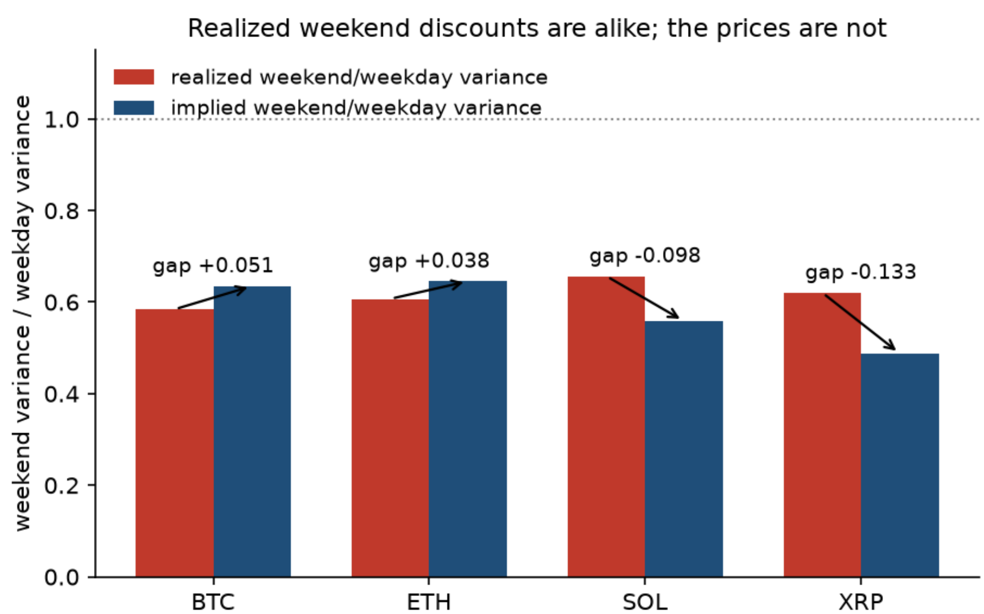

Figure 2. Implied against realized weekend variance ratios. Each asset's implied ratio with its confidence interval against its realized ratio; the 45-degree line is correct pricing.

5.2 The failure is within the weekend

Everything to this point concerns the level of the weekend discount, where the market does well. Its resolution is a separate question, and we find a sharper answer there.

Estimating a full day-of-week profile, with a separate implied effect for each of the seven days against the realized volatility of that day, we find the market's ordering otherwise good. Implied and realized profiles correlate at +0.98, +0.88, +0.89 and +0.82 across the four books, so the market knows Tuesday from Thursday. One place breaks, and it breaks the same way in every book.

Table 3. Saturday against Sunday, implied and realized.

	implied Saturday effect	*t*	realized Saturday effect
BTC	+0.004	0.14	−0.033
ETH	+0.074	2.01	−0.094
SOL	−0.012	−0.18	−0.075
XRP	+0.158	1.21	−0.041

Effect of a Saturday relative to a Sunday, in variance-ratio units. Positive implied values mean the market prices Saturday as the busier day.

In every asset the market prices Saturday as indistinguishable from, or busier than, Sunday, when Saturday is the quieter of the two. Not one of the four implied Saturday effects is significantly negative, while all four realized effects are negative. This is the one place in the weekly profile where we can show the market's ranking to be wrong rather than merely compressed, and it is the same place in all four books, across two settlement conventions, a decade of listing history and a factor of thirty in trade count.

We regard this as stronger evidence than the level results, for two reasons. Compression of a discount toward zero is what any smoothing or regularization in a pricing model would produce, and is consistent with the market knowing the answer and shading it. Ranking two adjacent days the wrong way round is not, since it requires the weekend to be represented as a single object. And because Saturday and Sunday sit inside the same weekend, an error between them cannot be attributed to anything that varies at weekly frequency, including the level of volatility, the news calendar and the funding environment, all of which are common to the pair.

We then ask why this error survives, and find the answer in the contract calendar rather than in beliefs. The natural response to a mispricing inside the weekend is to spread it: sell the Saturday-heavy contract and buy the Sunday-heavy one. Both legs sit inside the weekend, so the common weekend discount differences out and what remains is exactly the failure to rank the two days. We find that the trade cannot be put on. Expiries settle daily at 08:00 UTC, so which weekday an option's remaining life falls on is fixed entirely by its entry time and expiry date, and enumerating every hour of the week against every expiry in the half-day-to-eight-day window gives us not a sample of the menu a desk faces but the whole of it.

entry day	hours with any Sunday-heavy contract	hours offering **both** legs
Monday–Friday	**0 of 120**	**0 of 120**
Saturday	17 of 24	7 of 24
Sunday	21 of 24	6 of 24

A contract counts as tilted when its Saturday and Sunday shares differ by at least 15% of their sum.

Thirteen hours out of 168. From Monday to Friday nothing Sunday-heavy is listed at all, because any contract that reaches Sunday must pass through Saturday to get there. The two legs coexist only on Saturday between 07:00 and 13:00 UTC and on Sunday between 15:00 and 20:00, by which time the Saturday being sold is already under way or finished.

The market's one demonstrably wrong ranking of calendar time is therefore also the one its own contract calendar prevents anyone from arbitraging directly. We find this a more economical account of why this particular error survives than convention or inventory, and unlike those it requires no quote-level data, since it follows from the expiry schedule alone.

5.3 A decade-long trend, and what it tracks

The implied discount is not static, and Table 4 reports it by year.

Table 4. Implied weekend variance ratio, by year.

	2020	2021	2022	2023	2024	2025	2026
BTC	0.91	0.93	0.48	0.58	0.51	0.42	**0.30**
ETH	1.00	0.95	0.55	0.58	0.45	0.49	**0.47**

	2020	2021	2022	2023	2024	2025	2026
SOL	—	—	—	—	0.60	0.53	0.58
XRP	—	—	—	—	0.45	0.52	0.62

In 2020 Bitcoin and Ether priced essentially no weekend discount at all, against realized ratios of 0.52 and 0.70 in the same year, and by 2026 they price 0.30 and 0.47. Weighting each year by its own precision, the implied weekend effect falls by 0.144 per year in Bitcoin ($t = -12.5$) and 0.117 in Ether ($t = -10.5$). This also dissolves the apparent cross-sectional split in Table 2, since Solana and XRP quote deeper discounts than Bitcoin and Ether largely because their samples begin in 2024, late in a market-wide trend.

Measured as we measure it everywhere else in the paper, as mean weekend variance over mean weekday variance, the realized ratio has no significant trend over the same years ($t = -1.25$ and -1.09). The contrast looks like a market walking away from a benchmark that is standing still. We show that it is not, and the reason is a measurement point worth stating in general form.

A ratio of means of a right-skewed series is a weak estimator, and a weak estimator returning nothing is not evidence that nothing is there. Daily realized variance is extreme enough that a handful of days sets each year's mean. We use three diagnostics to separate an absent effect from an underpowered one, and all three agree here.

Table 5. The weekend trend, by moment of the realized distribution.

	ratio at mid-sample	per year	t	2020 → 2026
BTC arithmetic (ratio of means)	0.494	−0.062	−1.19	0.606 → 0.402
BTC **geometric** (ratio of centres)	0.437	**−0.136**	**−7.54**	0.685 → 0.279
ETH arithmetic	0.564	−0.037	−0.97	0.637 → 0.499
ETH **geometric**	0.532	**−0.098**	**−6.25**	0.736 → 0.384

Log points per year, within-month fixed effects, month-clustered standard errors.

We first trim. Cutting the top one per cent of days from each day type within each year, so that the trim itself carries no weekend effect, takes Bitcoin's arithmetic trend from −0.062 ($t = -1.2$) to −0.132 ($t = -4.9$), after which the estimate barely moves as the trim deepens to the median. The flatness was bought entirely in the extreme right tail.

We then refit at the centre. The same contrast on log variance estimates the ratio of geometric means and carries three to four times the *t*-statistic on the same days. The typical Bitcoin weekend has gone from carrying 69% of a typical weekday's variance to 28%.

We finally walk the sampling interval, which discriminates economics from measurement. Observation noise adds roughly the same amount to every day's measured variance, so it pushes a measured ratio toward one and attenuates any trend in it, hardest on the finest grid. A real trend seen through noise therefore strengthens as the interval coarsens, whereas a trend manufactured by noise that is itself shrinking weakens. Bitcoin's geometric trend runs −0.136 at five minutes and −0.193 at two hours, and Ether's runs −0.098 to −0.160, so it strengthens. Zero-return shares show no upward drift in either book, so we can rule out growing weekend staleness as the explanation.

Both realized measures are now falling, so the question becomes not whether the market responds to something but to which something. Fitting the implied ratio quarter by quarter gives trends of −0.186 ($t = -10.43$) in Bitcoin and −0.149 ($t = -10.46$) in Ether, against geometric realized trends of −0.136 and −0.098 and arithmetic ones of −0.062 and −0.037. Differencing implied against realized at each sampling interval, the *t*-statistic on the difference against the geometric benchmark falls monotonically to +0.2 and +0.4 at the coarsest sampling, where microstructure noise cannot reach, while against the arithmetic benchmark it sits near two standard errors throughout and closes only where power runs out. The implied trend converges on the geometric benchmark and never on the arithmetic one.

A backward-looking level test agrees. Regressing each quarter's log implied ratio on the previous quarter's realized ratio with asset fixed effects, which is what calibrating to recent history means, gives slopes of 0.71 on the geometric ratio and 0.27 on the arithmetic one. Both regressors are estimates, so both coefficients are attenuated, and correcting each by its own sampling variance lifts them to 1.15 and 0.48. The correction is large enough that we would read only the ranking off it, and the ranking is the one the trends give: the market's quotes move roughly one-for-one with where the weekend's variance usually sits, and about half as much with where its mean sits.

An option, however, pays off on expected total variance, which is an arithmetic mean. We therefore conclude that the market is tracking a real decline, at close to the right speed, in the wrong moment of the distribution. Section 6 shows that no proportional risk premium can move a variance ratio, so the mismatch is not repaired under the pricing measure. Its size is the gap between a weekend's typical variance and its average one, and that gap has been widening: measured as log mean minus mean log, a scale-free index of right-tail weight, the weekend's excess over the weekday's runs from −0.23 in 2020 to +0.31 in 2026 for Bitcoin and from −0.07 to +0.27 for Ether. The weekend has not simply gone quiet but has become lumpier, with ordinary weekends much quieter and rare weekends no less violent. A market that watches the middle of the distribution will mark the weekend down faster than its mean falls, by precisely the amount the tail is growing.

We note two limits on this rather than bury them. Solana and XRP show no trend on either side, which is consistent with a market-wide arc that began before they listed, but their standard errors are three to thirteen times the mature books' and their implied trends rest on nine or ten quarters. No difference test rejects anything for either, so they neither confirm the mechanism nor contradict it. And our comparison treats the option side and the bar side as independent when a volatile week moves both. The dependence is positive, which makes the reported standard errors conservative for the null that the two trends are equal and anti-conservative for the null that they differ.

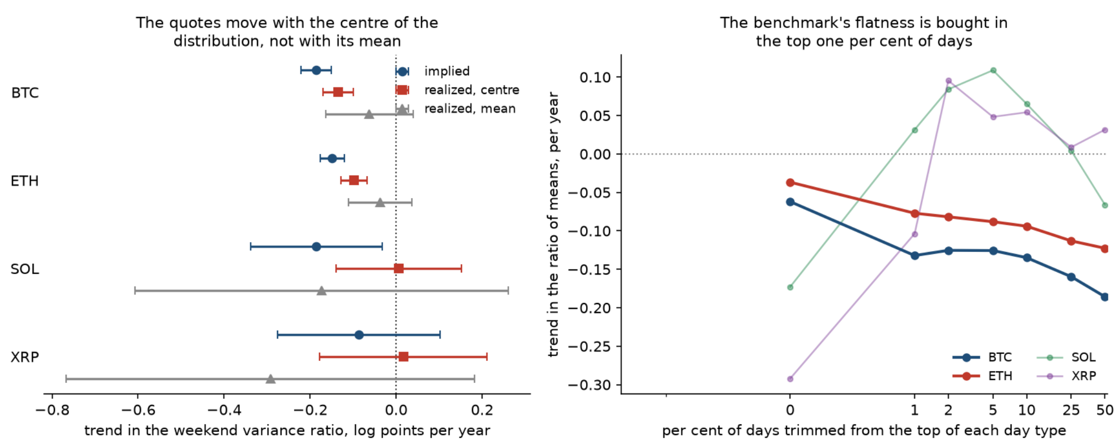

Figure 3. What the market tracks. Left: implied, realized-centre and realized-mean trends with 95% intervals, per asset. Right: the realized trend as the trimming fraction deepens.

6. Racing the risk-based explanation

Implied and realized quantities are measured under different probability measures, so a market pricing the weekend above its physical variance could be charging a risk premium rather than making an error. We take that possibility seriously enough to race it, and two features of our setting constrain what a premium is able to do.

The first is that a proportional premium cancels. If the risk-neutral measure inflates variance by a factor λ applied uniformly to calendar time, then both numerator and denominator scale by λ and the ratio we measure is unchanged. Only weekend-specific risk, something that differs in kind rather than merely in level, can move the quantity in Table 2. This is the constraint that makes the ratio formulation worth its loss of information relative to levels.

The second is that weekend tails are genuinely fatter. Standardized for scale, weekend returns carry 15% to 28% more mass beyond five standard deviations than weekday returns in all four books, so weekend-specific risk exists and has to be priced explicitly rather than argued away. We note that a companion tail asymmetry result on skew does not replicate, since two of the four books move the wrong way, and we therefore run the race on tail mass alone.

We bound what a jump premium can buy. Splitting realized variance in each regime into a continuous part and a jump part, and letting the risk-neutral measure price jump variance at a multiple $\kappa \geq 1$ of its physical value, the variance a dealer would charge is the continuous part plus κ times the jump part, and the priced weekend ratio $R^*(\kappa)$ is monotone in κ. Its limit as $\kappa \to \infty$ is therefore a bound, and no jump-risk premium of any size can push the priced ratio past it.

We use threshold truncation in the sense of Mancini (2009), at three standard deviations, taking the local scale from the same day's bipower variation (Barndorff-Nielsen and Shephard 2004) so that a quieter weekend is not mechanically classified as jump-free, and estimating an intraday seasonality factor separately for each regime in the manner of Andersen and Bollerslev (1997). Estimating the seasonality within regime matters here in a way it does not in most applications, because a factor pooled across day types would carry the very weekend effect we are trying to measure. Truncation misclassifies roughly 0.3% of ordinary returns by construction, though the bias applies to both regimes and largely cancels in the ratio. We check directly that the measured jump shares are not an artefact of the estimator's floor: run on a simulated pure diffusion of the same length and frequency, the same estimator returns a jump share under 5%, against the 24% to 36% these series produce.

Table 6. Residual gap under a jump-risk premium of size κ.

	κ = 1	κ = 3	κ = 10	bound ($\kappa \to \infty$)	reachable?
BTC	+0.051	+0.034	+0.019	**+0.009**	no
ETH	+0.039	+0.048	+0.056	**+0.039**	no
SOL	−0.098	−0.139	−0.184	**−0.098**	no
XRP	−0.136	−0.144	−0.150	**−0.136**	no

Implied ratio minus the ratio a market pricing weekend jump variance at κ times physical would quote. "Reachable" asks whether any κ closes the gap.

Jump compensation cannot account for the gap in any of the four books, and we find that it fails in three distinct ways. Bitcoin is the near miss: a premium does push its priced ratio toward the quote, but the whole path from $\kappa = 1$ to the bound closes only 82% of a gap of 0.051, and reaching even that far takes an unbounded premium. Ether fails on direction, since its weekend is less jump-intensive than its weekdays and every increase in κ widens the gap, from +0.039 at the physical measure to +0.056 at ten times it. Solana and XRP fail on sign, because their quotes sit below their realized ratios while a jump premium can only push a priced ratio up. For those two not even $\kappa = 0$ works: pricing weekend jump variance at zero still leaves continuous ratios of 0.616 and 0.613 against quotes of 0.558 and 0.488, so closing their gaps requires a negative price of jump risk, which is to say a market that pays to hold it.

Sampling uncertainty softens the verdict for any single asset without rescuing the story. Block-bootstrapping whole weeks and redrawing the implied ratio from its own sampling distribution, we find the probability that a jump premium of some size could reach the market's quote to be 0.27 for Bitcoin, 0.13 for Ether, 0.10 for Solana and 0.11 for XRP, none of which rejects at conventional levels on its own. What we regard as decisive is that the failure repeats in four books and requires opposite-signed premia in the two pairs, a large positive price of weekend jump risk in the mature books and a negative one in the two listed in 2024. No single risk premium does both. The result survives 44 estimator settings varying the truncation threshold, the seasonality treatment and the sampling frequency.

The option surface agrees, and disagrees with the premium story in the same direction. If weekend jump risk were being compensated, we would expect the compensation to concentrate where jump risk is priced, in the far wings. Estimating the weekend discount separately across the moneyness ladder, we find the far wing discounts the weekend harder than the money in all four books, by 0.10 to 0.13.

Fitted jointly so that the contrast carries a covariance, the far-wing-minus-at-the-money difference is −0.121 ($t = -4.15$) for Bitcoin and −0.118 ($t = -4.95$) for Ether, and equality across the smile is rejected outright in both ($\chi^2(3)$ = 18.4 and 28.4, $p < 0.001$). The two small books have the same sign and no precision. The at-the-money estimates also reproduce Table 2 almost exactly, at 0.630 against 0.635 for Bitcoin and 0.639 against 0.645 for Ether, which confirms that the widened sample has not changed the object being measured.

That the wings discount the weekend harder rather than more softly is the opposite of what weekend jump compensation implies. We read it instead as the smile's moneyness metric following the weekend clock only partway: a level adjustment made, and a structural one not.

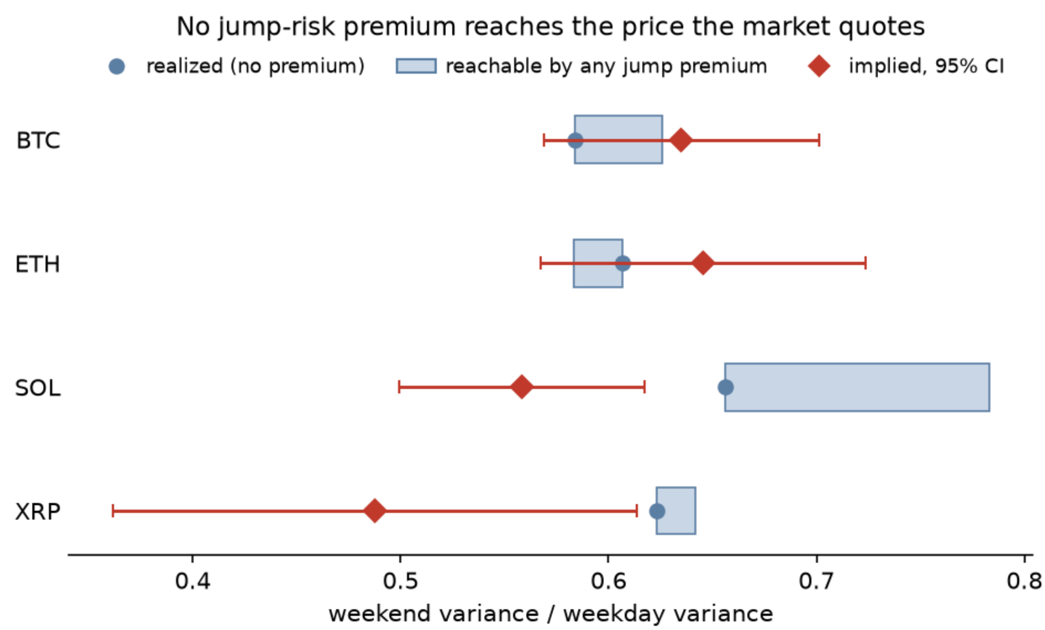

Figure 4. The horse race. Priced weekend variance ratio as the jump-risk premium κ grows, against each book's quote (diamond). The dashed line is the bound as $\kappa \to \infty$. Bitcoin approaches its quote and stops short; Ether moves away from it.

7. The error is measurable and has never been harvestable

A pricing error that no one can trade against invites the objection that it is an artefact of the estimator. We answer that objection in two steps: the error does show up in a hedged position, with the right sign and the right cross-sectional order, and it has nonetheless never cleared the venue's costs.

The trade that isolates the effect is a vega-matched calendar spread, in which we sell the weekend-heavy contract, buy the weekday-heavy one, delta-hedge both in the perpetual and hold to settlement.

We assign buckets within each day's own cross-section, because an absolute weekend-coverage threshold describes a spread whose two legs rarely exist at the same time. We measure costs rather than assume them, using Deribit's 0.03% option fee capped at 12.5% of premium, 0.05% taker on every perpetual rebalance, and an effective half-spread recovered from the tape by differencing buyer-paid against seller-received implied volatility on the same instrument-day. That last construction is the Roll (1984) intuition applied in volatility units, which the exchange-provided aggressor flag makes direct rather than inferential.

Table 7. The spread, gross and net of measured costs (daily rehedging).

per unit vega	BTC	ETH
gross spread	+0.0418	+0.0418
exchange and hedging fees, both legs	−0.0579	−0.0581
spread crossing, both legs	−0.0083	−0.0127
taker net	**−0.0245** (t = −2.09)	**−0.0289** (t = −1.82)
maker net (earns the spread rather than paying it)	−0.0079	−0.0036

The gross P&L validates the measurement chain. Ordering the four assets by the size of their measured pricing gap orders them by gross trading P&L as well, without exception, from Bitcoin's +0.051 gap and +0.023 gross down to XRP's −0.133 and −0.025. We regard this as a stronger check than a single profitable backtest, because the two legs differ in maturity as well as in weekend coverage, so a mechanical bias in which short-dated contracts out-earn longer-dated ones would generate same-signed P&L in every asset regardless of its pricing gap. Instead the sign and the rank track the gap across four books with two settlement conventions.

The signal is nonetheless smaller than the toll. Of the 0.066 per unit vega a Bitcoin taker pays, the bid-ask accounts for 0.008 and the remaining 0.058 is exchange and hedging fees, which fall on a maker exactly as they fall on a taker. Quoting rather than crossing is worth four half-spreads and closes roughly two thirds of the gap without reaching zero. Only a maker on a fee tier discounting option fees by 27% clears zero, at roughly +0.022 per unit vega, and not over the last twelve months, in which every construction we test loses money.

We conclude that the pricing error cannot be harvested by the trade that documents it, and that the obstacle is the venue's fee schedule rather than the market. This is consistent with the reading Section 5.3 suggests, in which a desk that prices the weekend clock correctly improves its marks on inventory it holds anyway, at no crossing cost and no capacity limit, and thereby leaves the quoted gap largely intact while making it unavailable to anyone who has to cross to reach it. We state that as an inference rather than a measurement, since separating it from the alternatives requires quote-level data this tape does not carry.

8. Conclusion

We find weekend variance in continuously traded cryptocurrency to be 34% to 42% below weekday variance on a venue that never closes, and roughly 65% below for a tokenized asset whose own underlying market is shut. Because trading is possible throughout, this isolates the information-arrival channel from the mechanical effect of closure, which is the separation French and Roll (1986) could not make and which Barclay, Litzenberger and Warner (1990) approached from the opposite direction by opening an exchange on Saturdays. Both designs point to trading time rather than calendar time as the relevant clock, and ours adds a dose-response: raise the amount of the traditional calendar that is switched off, and the weekend deepens on the same exchange under the same estimator.

The options market prices most of it. Roughly seven eighths of the realized weekend effect is in the price in both mature books, identified from contracts quoted at the same instant that differ only in how much weekend they span. That is a substantially better performance than the practitioner literature on calendar-time conventions would lead one to expect, and we think it worth stating plainly before the failures: this market reads the calendar well.

What it gets wrong is not the level but the resolution. All four books treat the weekend as one undifferentiated block, pricing Saturday as indistinguishable from Sunday when Saturday is reliably the quieter of the two, and because the two days sit inside the same weekend that error cannot be attributed to anything varying at weekly frequency. We then show that the venue's own expiry schedule makes this specific error unarbitrageable, offering both legs of the correcting spread in 13 hours of the week and never on a weekday. The market's one demonstrably wrong ranking of calendar time is the one its own contract calendar protects. We find this a more economical account of why an error persists than convention or inventory, and unlike those it follows from the listing schedule alone.

The implied discount has meanwhile deepened for a decade, at 0.144 and 0.117 per year, and we show that trend to be a response to a real change in the data measured against the wrong moment of it. The typical weekend really has been getting quieter, at 0.136 log points a year at the centre of Bitcoin's distribution, against an arithmetic benchmark that appears flat only because a handful of days sets each year's mean. The market tracks the centre while an option pays off on the mean, and since Section 6 rules out any proportional risk premium moving a variance ratio, the mismatch is not repaired under the pricing measure. Its size is the widening gap between a weekend's typical variance and its average one.

That last result carries a methodological point beyond this application. A ratio of means of a right-skewed series is a weak estimator, and a weak estimator returning nothing is not evidence that nothing is there. Three cheap diagnostics separate the two, namely trimming the tail and refitting, refitting at the centre of the distribution, and walking the sampling interval, and here all three agree that a non-rejection was a power failure. Where a non-rejection is load-bearing, we think it warrants the scrutiny a rejection would receive.

Three limitations bound our claims. Solana and XRP carry standard errors three to thirteen times the mature books', so they replicate the central realized fact without sharpening any of the pricing tests, and we have been careful not to lean on them. The gold result is a lower bound, because PAXG's weekend staleness biases its measured ratio toward one. And our identification assumes the expiry schedule is exogenous to weekend variance, which we treat as a maintained assumption, though a listing calendar fixed years in advance and uniform across underlyings makes it a mild one.

What we cannot say from a trade tape is why a desk would calibrate to the centre of a right-skewed distribution when the payoff is on its mean. Robust estimation of a skewed quantity is ordinary practice and defensible on its own terms; it is simply the wrong practice for this quantity. Separating deliberate robustness from a fitted model with a thin-tailed error, or from nobody looking, requires quote-level data and a view of who is on the other side of the trade.

Electronic companion

The e-companion contains the full identification appendix and within-day binscatters for all four assets; the pooled convention test and its power analysis; robustness of the realized effect across sampling frequencies, jump truncation thresholds and seasonality treatments; the complete day-of-week profile tables; the full trimming and sampling ladders behind Table 5; the 44-setting robustness grid behind Table 6; the smile and wing analysis; and the complete trading results, including the rehedge ladder, the outright-short variant, the contract-selection sweep and the maker fee decomposition.

A full replication package accompanies the submission, containing all collection and estimation code, a mapping from every table, figure and in-text number to the script that produces it, and a test suite that pins each estimator against simulated data with a planted answer. We use no licensed or proprietary data, and every input comes from a public API requiring no account or agreement, so the paper is reproducible from nothing.

References

Alexander C, Imeraj A (2023) Delta hedging bitcoin options with a smile. *Quant. Finance* 23(5):799–817.

Andersen TG, Bollerslev T (1997) Intraday periodicity and volatility persistence in financial markets. *J. Empirical Finance* 4(2–3):115–158.

Barclay MJ, Litzenberger RH, Warner JB (1990) Private information, trading volume, and stock-return variances. *Rev. Financial Stud.* 3(2):233–253.

Barndorff-Nielsen OE, Shephard N (2004) Power and bipower variation with stochastic volatility and jumps. *J. Financial Econometrics* 2(1):1–37.

Bollen NPB, Whaley RE (2004) Does net buying pressure affect the shape of implied volatility functions? *J. Finance* 59(2):711–753.

Carr P, Wu L (2009) Variance risk premiums. *Rev. Financial Stud.* 22(3):1311–1341.

French KR, Roll R (1986) Stock return variances: The arrival of information and the reaction of traders. *J. Financial Econom.* 17(1):5–26.

Gârleanu N, Pedersen LH, Poteshman AM (2009) Demand-based option pricing. *Rev. Financial Stud.* 22(10):4259–4299.

Hansen PR, Lunde A (2006) Realized variance and market microstructure noise. *J. Bus. Econom. Statist.* 24(2):127–161.

Mancini C (2009) Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scand. J. Statist.* 36(2):270–296.

Oldfield GS, Rogalski RJ (1980) A theory of common stock returns over trading and non-trading periods. *J. Finance* 35(3):729–751.

Roll R (1984) A simple implicit measure of the effective bid-ask spread in an efficient market. *J. Finance* 39(4):1127–1139.

Zhang L, Mykland PA, Aït-Sahalia Y (2005) A tale of two time scales: Determining integrated volatility with noisy high-frequency data. *J. Amer. Statist. Assoc.* 100(472):1394–1411.